Sim2Real Ablation Matrix

Vision-based RL on a WidowX AI arm (Lift-Cube, Push-T). Top: which ablations to run first, from each factor's influence on the results, what is already known and the cost.
Below: every option rated on how well it does (first four columns) and what it costs (last two). Solid border = measured, dashed = estimated.

ABLATION PRIORITIES

	PRIORITY	SIM2REAL influence	SIM RESULTS influence	KNOWN?	COST to ablate	WHY
Encoder	1 · Must ablate	High	High	Partly	High	Main question of the thesis. So far only compared in sim, not on the real robot.
Table texture randomization	1 · Must ablate	High	High	Partly	Mid	Probably depends on the encoder. Only tested with DINOv2-AFA so far.
Backbone training	1 · Must ablate	High	High	Open	Mid	Fine-tuning should be the upper bound, but it was never trained.
Lighting randomization	2 · Should ablate	High	Mid	Partly	Low	Wrong brightness already broke one policy on the robot. The range width is not tested yet.
Object and goal colours	2 · Should ablate	Mid	Mid	Open	Low	Probably interacts with the training scene, so it can run together with it.
Backbone size	2 · Should ablate	Mid	Mid	Open	Mid	Shows if a bigger pretrained model transfers better or only costs more.
Dynamics randomization	3 · Optional or settled	High	Mid	Open	Low	Matters for transfer, but the same for every encoder. General setup, not part of the encoder study.
Smoothness penalties	3 · Optional or settled	High	Mid	Partly	Low	Needed for smooth motion on the robot. Same for every encoder, so it is part of the setup.
Dynamics model	3 · Optional or settled	High	Mid	Settled	Mid	The measured arm already matches the real joint traces.
Camera exposure and white balance	3 · Optional or settled	High	Low	Settled	Low	Has to be set right before every session. Not something to ablate.
Camera pose calibration	3 · Optional or settled	High	Low	Settled	Low	Calibrated once with the ChArUco board. The small jitter stays in training.
Action type	3 · Optional or settled	Mid	High	Settled	Mid	Target-relative with the target in the observation worked best in sim.
PPO settings	3 · Optional or settled	Mid	High	Settled	Low	Good values are known. They should stay fixed for all encoder runs.
Contact penalties	3 · Optional or settled	Mid	High	Settled	Low	Penalizing contact with the T always made the policy stop touching it.
Goal signal	3 · Optional or settled	Mid	Mid	Settled	Low	Giving the goal pose as input worked better in sim, so it stays.
Feature-map resolution	3 · Optional or settled	Mid	Mid	Partly	Low	R3M layer3 vs layer4 already shows that finer features help.
Goal marker shape	3 · Optional or settled	Mid	Mid	Partly	Low	The outline is used now. No clean comparison exists and there is not much to gain.
Control delay in sim	3 · Optional or settled	Mid	Low	Settled	Low	Measured on the robot at 25 to 55 ms and randomized in training now. No ablation needed.
Proprioception noise	3 · Optional or settled	Mid	Low	Open	Low	Easy to add, but the effect is probably small.
Control rate	3 · Optional or settled	Low	High	Partly	Low	20 Hz did better in sim, but it also sees more and shorter episodes. Same for every encoder, so part of the setup.
Input image resolution	3 · Optional or settled	Low	Mid	Open	Mid	Mostly a question of compute.
Feature cache precision	3 · Optional or settled	Low	Low	Settled	Low	fp32 on the robot gave almost the same success as bf16 training.
Latent size (encoder output)	3 · Optional or settled	Low	Low	Open	Low	Probably does not matter much.

OPTIONS IN DETAIL

PRIORITY 1 · MUST ABLATE

	SIM2REAL transfer	SIM SUCCESS	SAMPLE efficiency	ENCODING efficiency	WALL CLOCK cost	VRAM cost
ENCODER						
Local pretrained (DINOv2 + AFA)	High	High	High	High	Mid	Mid
Global pretrained (DINOv2 / R3M + linear)	Mid	Low	Low	Low	Mid	Low
Scratch (NatureCNN / CompactViT)	Low	Mid	Low	Mid	Low	Mid
BACKBONE TRAINING						
Frozen pretrained backbone (≈ 0.1 M trainable in AFA)	High	Mid	High	High	Mid	Mid
Fine-tuned pretrained backbone (≈ 21 M trainable)	High	High	Mid	High	High	High
Trained from scratch (≈ 0.1–1 M trainable)	Low	Mid	Low	Mid	Low	Mid
TABLE TEXTURE RANDOMIZATION						
Randomized table textures (1,203 photos)	High	Mid	Low	Low	Low	Low
One fixed photo of the real table (rig scene)	Mid	High	High	High	Low	Low

PRIORITY 2 · SHOULD ABLATE

	SIM2REAL transfer	SIM SUCCESS	SAMPLE efficiency	ENCODING efficiency	WALL CLOCK cost	VRAM cost
LIGHTING RANDOMIZATION						
Matched to the rig with margin (current)	High	High	High	High	Low	Low
Wide random lighting (intensity, colour, direction)	High	Mid	Mid	Mid	Low	Low
Fixed lighting	Low	High	High	High	Low	Low
BACKBONE SIZE						
DINOv2 ViT-B/14 (86 M)	High	High	High	High	High	High
DINOv2 ViT-S/14 (21 M, current)	Mid	High	High	High	Mid	Mid
OBJECT AND GOAL COLOURS						
Narrow random range around the rig's colours	High	High	High	High	Low	Low
Random colours (full RGB)	High	Mid	Low	Low	Low	Low
Exactly fixed colours	Low	High	High	High	Low	Low

PRIORITY 3 · OPTIONAL OR SETTLED

	SIM2REAL transfer	SIM SUCCESS	SAMPLE efficiency	ENCODING efficiency	WALL CLOCK cost	VRAM cost
DYNAMICS MODEL						
Measured arm, object mass and friction	High	Mid	Mid	Mid	Low	Low
Default MJCF values	Low	High	High	Mid	Low	Low
DYNAMICS RANDOMIZATION						
Measured model + randomized gains, damping, mass, friction	High	Mid	Low	Mid	Low	Low
Measured model, only friction randomized	Mid	High	High	Mid	Low	Low
SMOOTHNESS PENALTIES						
Action-rate and near-goal speed penalty	High	Mid	Low	Mid	Low	Low
Weak or no smoothness penalty	Low	High	High	Mid	Low	Low
CAMERA EXPOSURE AND WHITE BALANCE						
Matched to the training brightness	High	Mid	Mid	High	Low	Low
Camera defaults	Low	Mid	Mid	Low	Low	Low
CAMERA POSE CALIBRATION						
Calibrated, with small per-step pose and FOV jitter (current)	High	Mid	Mid	Mid	Low	Low
Calibrated, fixed	Mid	High	High	High	Low	Low
Uncalibrated (hand-placed, nominal FOV)	Low	High	High	High	Low	Low
ACTION TYPE						
Target-relative delta (target observed)	High	High	High	Mid	Low	Low
Measured-relative delta	Mid	Mid	Mid	Mid	Low	Low
Absolute joint targets	Mid	Low	Low	Mid	Low	Low
PPO SETTINGS						
Tuned (KL 0.05, entropy 0.001, std floor)	Mid	High	High	Mid	Low	Low
Loose (KL 0.2, no entropy, no floor)	Low	Mid	Mid	Low	Low	Low
CONTACT PENALTIES						
Small penalty on pressing the T into the table	Mid	Mid	Mid	Mid	Low	Low
No contact penalties	Low	High	High	Mid	Low	Low
Penalty or termination on gripper–T force	Low	Low	Low	Mid	Low	Low
FEATURE-MAP RESOLUTION						
16 × 16 or 14 × 14 tokens (DINOv2 patch 14, R3M layer3)	Mid	High	High	High	Mid	High
7 × 7 cells (R3M layer4)	Mid	Mid	Low	Low	Low	Mid
GOAL SIGNAL						
Goal pose as input + marker	Mid	High	High	High	Low	Low
Marker in the image only	Low	Mid	Mid	Mid	Low	Low
GOAL MARKER SHAPE						
Hollow outline	High	High	High	High	Low	Low
Filled T	Low	Mid	Mid	Low	Low	Low
CONTROL DELAY IN SIM						
Randomized around the measured 25–55 ms (current)	High	Mid	Mid	Mid	Low	Low
Almost no delay (0–5 ms, before)	Mid	High	High	Mid	Low	Low
PROPRIOCEPTION NOISE						
Noise on joint positions and velocities in training	Mid	Mid	Mid	Mid	Low	Low
Exact proprioception	Low	High	High	Mid	Low	Low
CONTROL RATE						
20 Hz	Mid	High	High	Mid	Low	Low
50 Hz	Mid	Mid	Mid	Mid	Low	Low
INPUT IMAGE RESOLUTION						
224 × 224 px	Mid	High	High	High	Mid	Mid
128 × 128 px	Mid	Mid	Mid	Mid	Low	Low
FEATURE CACHE PRECISION						
bf16 cache (current)	Mid	Mid	Mid	Mid	Low	Low
fp32 cache	Mid	Mid	Mid	Mid	Mid	Mid
LATENT SIZE (ENCODER OUTPUT)						
256 (current)	Mid	High	High	High	Low	Low
Large (1024)	Mid	High	Mid	Mid	Low	Low
Small (64)	Mid	Mid	Mid	Mid	Low	Low